

## State Space Analysis

There are different methods of representation of a control system like Block diagram, Signal flow graphs, etc. In all the above methods the dynamics of the system is represented by an I/P - O/P relation called Transfer function, however these approaches have some limitations. They are

- ① The initial conditions of system were not taken into account
- ② The method is insufficient/troublesome to give complete time domain soln of higher order system.
- ③ It is not convenient for analysis of MIMO systems.
- ④ It gives analysis of system for specific types of inputs like step, ramp etc.
- ⑤ It is applicable only to linear time invariant systems.

(Hence, it is absolutely necessary to use a method of analyzing systems, which overcomes most of the above said difficulties)

Therefore the modern method known as "State Space analysis" is used to overcome these limitations. This method represents the past, present, & future status of the system & also helps to represent the  $n^{\text{th}}$  order diff Eq<sup>n</sup> as 'n' no of 1<sup>st</sup> order differential Equations. This makes the calculation easier.

## State space Modeling

- 1) Introduction to state space modelling and.
  - \* Definition
  - \* How it relates to other modeling forms like Differential eq<sup>n</sup>, TF, BPR, SFG etc.
- 2) Choosing state variables.
- 3) Transforming between model types.
- 4) State-space modelling example.

## State space Modeling

- A State space model represents a system by a series of 1st order differential state equations and algebraic o/p equations }  
 i.e.  $n^{\text{th}}$  order differential eq<sup>n</sup> is rearranged as series of 1st order differential equations.
- The reason we use 1<sup>st</sup> order, is.  
 They are numerically efficient to solve, they can handle more complex systems, allow for more geometric understanding of dynamic systems, and form the basis for modern control theory.

## Example

Consider 3<sup>rd</sup> order differential equation, where  $u(t)$  is the input and  $\dot{x}(t)$  is the output.



$$\ddot{x} + 5\dot{x} + 3x + 2x = u.$$

By definition this is not in state space form,  
In order to be in state space form, we need to  
model this system as series of 1st order differential equations.  
Here the highest derivative is 3rd order.

we can generate a state space model by pure  
mathematical manipulation through changing variables.

By choosing these new set of variables.

$x_1 = x$  (0th derivative),  $x_2 = \dot{x}$  (1st derivative),  $x_3 = \ddot{x}$  (2nd derivative)  
we can rearrange this as 3 1st order differential equations.  
(The highest derivative does not need state variable)

For each of the state variables, we are going to have  
a single differential equation, so our 1st state variable.

State Equations

$$\begin{cases} \dot{x}_1 = \dot{x} = x_2 \rightarrow (1) \\ \dot{x}_2 = \ddot{x} = x_3 \rightarrow (2) \\ \dot{x}_3 = \dddot{x} = u - 5\dot{x} - 3\ddot{x} - 2x \rightarrow (3) \end{cases}$$

$$\dot{x}_3 = \ddot{x} = -5x_3 - 3x_2 - 2x_1 + u$$

what we have done is we have taken our original  
3rd order differential eqn model and written it as three  
1st order differential equations.

we define these 3 equations as our state  
Equations.

State Space form also includes an output equation or a series of o/p equations. (2)

What defines the output is sort of an engineering decision you know it depends what ~~we~~ <sup>we are</sup> trying to control, it depends on what we are trying to analyze.  
 Eg: position can be o/p or velocity can be o/p or both can be o/p.

So let's say arbitrarily that in this case  $\dot{x}$  is the output (ie. velocity).

Output Equation  $\left\{ \begin{array}{l} y = \dot{x} = x_2 \end{array} \right.$

→ (4)

State Eq<sup>n</sup>'s + O/p Eq<sup>n</sup>'s ⇒ State Space model ⇒ used to model Dynamic System.

System has 1 P/p (u)  
 1 O/p (y)  
 & 3 State variables ( $x_1, x_2, x_3$ )

For linear systems, we can write state model as

Matrices

(vector of derivative of state variables)  $\dot{\vec{x}} = A \cdot \vec{x} + B \cdot \vec{u}$

(vector of o/p's)  $\vec{y} = C \cdot \vec{x} + D \cdot \vec{u}$

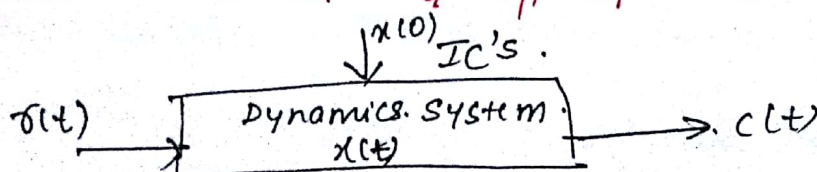
$A$  = system matrix.  $B$  = Input Matrix.

$C$  = o/p<sup>n</sup> Matrix.  $D$  = feed-forward Matrix & (transition matrix).

$\vec{x} \Rightarrow$  state vector of state variables

$\vec{u} \Rightarrow$  vector of i/p variables (coupling/control)

Let's Rewrite state Eq<sup>n</sup>'s & o/p eq<sup>n</sup>s in matrix form.



D ⇒ Will come if System TF is not in proper form



State Eq<sup>n</sup>

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -3 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

o/p Eq<sup>n</sup>

$$[y] = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} u$$

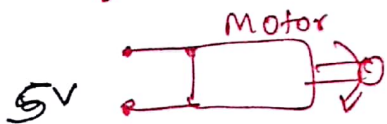
(If we blindly do this change of variables we may end up with a situation where we are using more state variables that are not necessary, we will be using larger set of Eq<sup>n</sup> than we need to)

Choosing State Variables

i/p → System → o/p

Static system :- current o/p depends only on current i/p (relation will be algebraic)

(eg: LR Dynamic)



Apply 5V to motor that does not tell us the speed of motor. Specifically motor will behave differently

if. right before we applied 5V, we had applied 0V or right before we applied 5V, we had applied 100V

That past, affects the current behaviour of the system.

When we change the i/p to 5V, the o/p does not change instantaneously & it takes time to react. & there is

inertia. , So with a dynamic system, we can define

its state and the state of the system basically

captures that history, it captures all of those past i/p's.

Question:  $\rightarrow$  what initial conditions do I need to capture the system's state? (3)

What set of variables will capture all of the history that came before this instant, with that in mind we define some terms, we are going to define the state of a dynamic system as being captured by a set of variables and if we know the values of these variables then we know the state of the system.

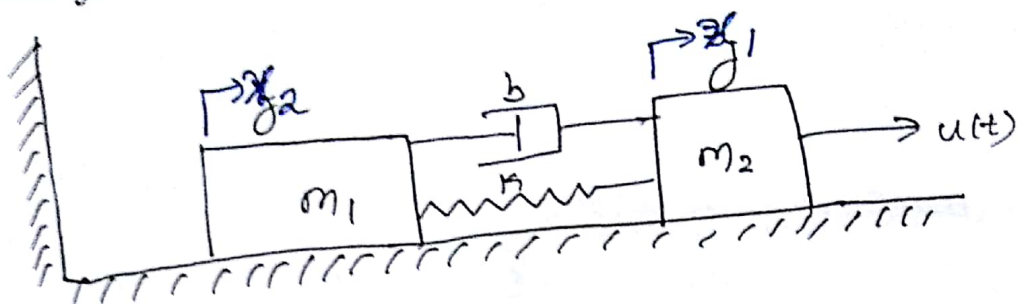
And if we know the state of the system at a current instant of time then knowledge of future i/p's will completely allow us to determine the behavior of the system. This sort of capturing of state captures everything that happened in the past and therefore knowledge of i/p allow us to predict what's going to happen in the future.

~~Knowledge of i/p's allows us to p.~~  
The minimum no. of state variables that we need.  
the minimum no. of variables to capture a system  
state is equal to the no. of independent energy  
storage elements.

Energy stored in the system at a current instant of time affects the behavior of the system going forward. and so if we can capture/determine the energy that is stored, then we can determine what's going to happen in future?



Consider this eq:



Equations of motion

for  $x_2$   $m_1 \ddot{x}_2 + b(\dot{x}_2 - \dot{x}_1) + k(x_2 - x_1) = 0$

for  $x_1$   $m_2 \ddot{x}_1 + b(\dot{x}_1 - \dot{x}_2) + k(x_1 - x_2) = u(t)$

One <sup>choice</sup> of state variables

$$\left. \begin{aligned} x_1 &= y_1 \\ x_2 &= \dot{y}_1 \\ x_3 &= y_2 \\ x_4 &= \dot{y}_2 \end{aligned} \right\}$$

We have 2 2<sup>nd</sup> order differential eq<sup>n</sup>s  
we can transform them into 4, 1<sup>st</sup> order  
differential eq<sup>n</sup>, Just by using change  
of variable.

But it turns out that this set of  
state variables is not minimum. we can put our  
system in to state space form with a smaller no. of  
state variables than the ones given here

The way to do this is to identify where  
energy is stored in the system

eg: In this ckt, Energy is stored in mass & spring.

Spring  $\Rightarrow$  stores PE (potential energy)

We need a variable that captures the energy stored  
in spring, Here energy stored in spring is determined by  
its amount of deformation.

deformation is determined by  $\Rightarrow y_2 - y_1$

$\therefore x_1 = (y_2 - y_1) \rightarrow (1)$  *(we need not know  $y_2, y_1$  individually it's enough to know difference)*

There is KE (kinetic energy) stored in mass.

$x_2 = \dot{y}_2$  (determined by velocity)  $\rightarrow (2)$

$x_3 = \dot{y}_1$   $\rightarrow (3)$

These 3 state variables captures the state of system.

Damper (Dashpot  $\Rightarrow$  don't store energy, they dissipate energy.)

Write 1<sup>st</sup> derivatives for these 3 states.

$$\begin{cases} x_1 = (y_2 - y_1) \Rightarrow \dot{x}_1 = x_2 - x_3 \rightarrow (1) \\ x_2 = \dot{y}_2 \Rightarrow \dot{x}_2 = \ddot{y}_2 = \frac{1}{m_1} (-b(x_2 - x_3) - k(x_1)) \rightarrow (2) \\ x_3 = \dot{y}_1 \Rightarrow \dot{x}_3 = \ddot{y}_1 = \frac{1}{m_2} (-b(x_3 - x_2) + kx_1 + u) \rightarrow (3) \end{cases}$$

2<sup>nd</sup> order differential Eq<sup>n</sup>'s are written as (3)  
1<sup>st</sup> order differential Eq<sup>n</sup>'s

lets put in matrix form.

$$\underbrace{\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix}}_{\dot{\vec{x}}} = \underbrace{\begin{bmatrix} 0 & 1 & -1 \\ -\frac{k}{m_1} & -\frac{b}{m_1} & \frac{b}{m_1} \\ \frac{k}{m_2} & \frac{b}{m_2} & -\frac{b}{m_2} \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\vec{x}} + \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1/m_2 \end{bmatrix}}_B u$$

We also have o/p Eq<sup>n</sup>.

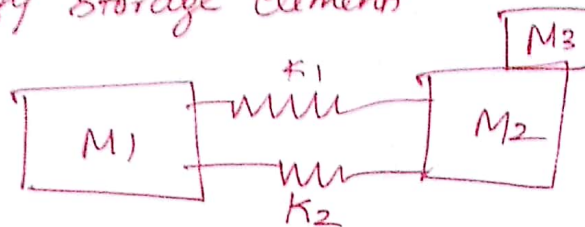
let o/p =  $y = \dot{y}_2$

$$y = \underbrace{\begin{bmatrix} 0 & 1 & 0 \end{bmatrix}}_C \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_D + \underbrace{\begin{bmatrix} 0 \end{bmatrix}}_D u$$



Note: ① If out o/p is  $y_2 \Rightarrow$  then I can't use  $(y_2 - y_1)$  state, I have to use 2 states  $\rightarrow y_2 \rightarrow y_1$ .

② If components are dependent then they don't need their own energy storage elements

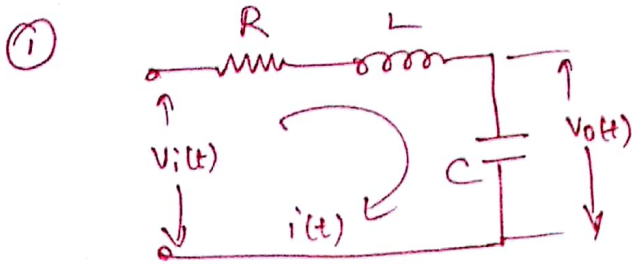


$K_1, K_2$  are dependent.  
 $M_3$  &  $M_2$  are dependent.  
(they have same displacement)

## State space Numericals

①

① Obtain the state model for the following electrical networks.



There are 2 energy storing elements  $L$  &  $C$ , so min 2 state variables are required.

$x_1(t) = i(t) =$  current through inductor.

$x_2(t) = v_o(t) =$  voltage across the capacitor.

Writing integro diff eqn we have.

$$R \cdot i(t) + L \cdot \frac{di(t)}{dt} + \underbrace{\frac{1}{C} \int i(t) dt}_{v_o(t)} = v_i(t) \rightarrow \textcircled{1}$$

$$\& \int v_o(t) = \frac{1}{C} \int i(t) \cdot dt = x_2(t) \Rightarrow \boxed{\dot{x}_2(t) = \frac{1}{C} i(t) = \frac{1}{C} x_1(t)} \rightarrow \textcircled{2}$$

So on eqn ①. substitute.

$$x_1(t) = i(t) \quad \& \quad x_2(t) = \frac{1}{C} \int i(t) \cdot dt = v_o(t)$$

Then we get -

$$R \cdot x_1(t) + L \cdot \dot{x}_1(t) + x_2(t) = v_i(t)$$

$$\therefore \boxed{\begin{aligned} \dot{x}_1(t) &= -\frac{R}{L} x_1(t) - \frac{1}{L} x_2(t) + \frac{v_i(t)}{L} \\ \& \quad \dot{x}_2(t) &= \frac{1}{C} x_1(t) \end{aligned}}$$

Arranging the above eqn in matrix form.

$$\dot{\vec{X}} = A\vec{X} + B\vec{U} \quad \text{we get.}$$

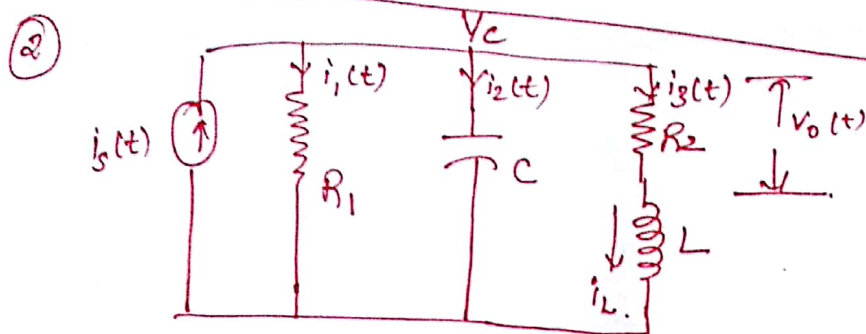
State matrix  $\rightarrow$

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & -\frac{1}{L} \\ \frac{1}{C} & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} v_i(t)$$

$A$   $B$   $u$



o/p eqn.  $y = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$  where  $D = 0$ .



Let  $x_1(t) = v_c(t)$  &  $x_2(t) = i_3(t) = i_L(t)$   
 Apply KCL.

$$i_s(t) = \frac{v_c}{R_1} + C \cdot \frac{dv_c}{dt} + i_L \rightarrow (1)$$

$$v_c(t) = R_2 i_L + L \cdot \frac{di_L}{dt} \rightarrow (2)$$

from (1)

$$i_s(t) = \frac{\dot{x}_1(t)}{R_1} + C \cdot \dot{x}_1(t) + x_2(t)$$

$$\Rightarrow \dot{x}_1(t) = -\frac{1}{R_1 C} x_1(t) - \frac{1}{C} x_2(t) + \frac{i_s(t)}{C}$$

from (2)

$$v_c(t) = R_2 x_2(t) + L \cdot \dot{x}_2(t)$$

$$\Rightarrow \dot{x}_2(t) = \frac{x_1(t)}{L} - \frac{R_2}{L} x_2(t)$$

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_1 C} & -\frac{1}{C} \\ \frac{1}{L} & -\frac{R_2}{L} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 1/C \\ 0 \end{bmatrix} [i_s(t)]$$

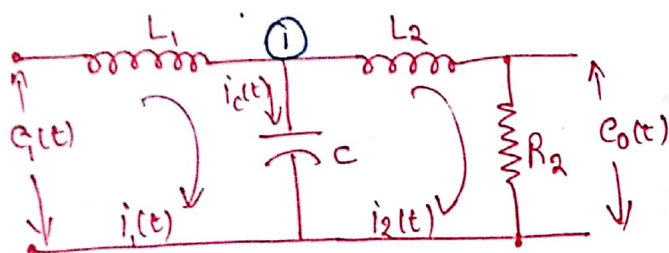
1st Eqn

$$Y = CX + DU$$

$$V_o(t) = i_2(t) \cdot R_2 = R_2 x_2(t)$$

$$Y = \begin{bmatrix} 0 & R_2 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \frac{D}{0} U$$

③



$$\text{Let } x_1(t) = i_1(t)$$

$$x_2(t) = i_2(t)$$

$$x_3(t) = V_c(t)$$

Note: These variables can be interchanged, Thus the state model is not the unique property of the system.

Apply KVL to both the loops & KCL at node ①.

(KVL L1) 
$$V_i(t) = L_1 \frac{di_1(t)}{dt} + \frac{1}{C} \int (i_1(t) - i_2(t)) dt = L_1 \frac{di_1(t)}{dt} + V_c(t)$$

$$\therefore \boxed{\dot{x}_1(t) = \frac{V_i(t)}{L_1} - \frac{x_3(t)}{L_1}} \rightarrow \text{①}$$

(KVL L2) 
$$0 = \frac{1}{C} \int (i_2 - i_1) dt + L_2 \frac{di_2}{dt} + R_2 i_2$$

$$0 = -V_c(t) + L_2 \dot{x}_2(t) + R_2 x_2(t)$$

$$\boxed{\dot{x}_2(t) = \frac{R_2}{L_2} x_2(t) + \frac{x_3(t)}{L_2}} \rightarrow \text{②}$$

(KCL ①)

$$i_1(t) = i_c(t) + i_2(t) = C \frac{dV_c}{dt} + i_2(t)$$

$$x_1(t) = C \dot{x}_3(t) + x_2(t)$$



$$\therefore \dot{x}_3(t) = \frac{x_1(t)}{C} - \frac{x_2(t)}{C} \rightarrow (3)$$

State  
Matrix  
Eqn.

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 0 & -\frac{1}{L} \\ 0 & -\frac{R_2}{L_2} & \frac{1}{L_2} \\ \frac{1}{C} & -\frac{1}{C} & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 1/4 \\ 0 \\ 0 \end{bmatrix} e_i(t)$$

O/p Eqn

$$Y = e_o(t) = i_2 R_2$$

$$Y = \begin{bmatrix} 0 & R_2 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$$

(4) Write the state equations for the following.

$$4 \cdot \frac{d^3 y(t)}{dt^3} + 2 \cdot \frac{d^2 y(t)}{dt^2} + \frac{dy(t)}{dt} + 2y(t) = 5 \cdot x(t)$$

(The order of the system determine the no. of variable & size of A matrix  $n=3$ ).

$$4 \ddot{y}(t) + 2 \ddot{y}(t) + \dot{y}(t) + 2y(t) = 5 \cdot x(t)$$

Let.

$$y(t) = x_1(t)$$

$$\dot{y}(t) = \dot{x}_1(t) = x_2 \rightarrow (1)$$

$$\ddot{y}(t) = \ddot{x}_1(t) = \dot{x}_2(t) = x_3(t) \rightarrow (2)$$

$$\ddot{y}(t) = \ddot{x}_1(t) = \ddot{x}_2(t) = \dot{x}_3(t) = ? \quad \text{etc}$$

$$\therefore 4\ddot{y}(t) + 2\dot{y}(t) + \dot{y}(t) + 2y(t) = 5 \cdot \delta(t)$$

$$4 \cdot \ddot{x}_3(t) + 2 \dot{x}_3(t) + x_2(t) + 2 \cdot x_1(t) = 5 \cdot r(t)$$

$$\dot{x}_3(t) = -\frac{1}{2}x_3(t) - \frac{1}{4}x_2(t) - \frac{1}{2}x_1(t) + \frac{5}{4}r(t)$$

$$\therefore \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\frac{1}{2} & -\frac{1}{4} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 5/4 \end{bmatrix} r(t)$$

5) Given  $\frac{C(s)}{R(s)} = \frac{10}{s^4 + 4s^3 + 2s^2 + s + 2}$

Soln.  $(s^4 + 4s^3 + 2s^2 + s + 2) C(s) = 10 \cdot R(s)$

Wkt  $\mathcal{L}\left\{\frac{d}{dt}c(t)\right\} = s \cdot C(s) \quad \therefore \mathcal{L}^{-1}\{s \cdot C(s)\} = \frac{d}{dt}c(t)$   
 $\mathcal{L}\left\{\frac{d^2}{dt^2}c(t)\right\} = s^2 C(s) \quad \mathcal{L}^{-1}\{s^2 C(s)\} = \frac{d^2}{dt^2}c(t) \quad \text{etc.}$

$$\frac{d^4 c(t)}{dt^4} + 4 \frac{d^3 c(t)}{dt^3} + 2 \frac{d^2 c(t)}{dt^2} + \frac{d c(t)}{dt} + 2 \cdot c(t) = 10 \cdot \delta(t)$$

Let  
 $c(t) \Rightarrow x_1(t)$   
 $\dot{c}(t) = \dot{x}_1(t) = x_2(t)$   
 $\ddot{c}(t) = \dot{x}_2(t) = x_3(t)$   
 $\dddot{c}(t) = \dot{x}_3(t) = x_4(t)$   
 $c^{(4)}(t) = \dot{x}_4(t)$

$$\therefore \dot{x}_4(t) + 4 \cdot x_4(t) + 2 \cdot x_3(t) + x_2(t) + 2 \cdot x_1(t) = 10 \cdot \delta(t)$$



$$\dot{x}_4(t) = -2 \cdot x_1(t) - x_2(t) - 2 \cdot x_3(t) - 4 \cdot x_4(t) + 10 \cdot r(t)$$

$$\begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \\ \dot{x}_3(t) \\ \dot{x}_4(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -2 & -1 & -2 & -4 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \\ x_4(t) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 10 \end{bmatrix} r(t)$$

7) Obtain the DE for the following state Eqn.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -\frac{1}{3} & -\frac{5}{3} & -\frac{3}{4} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{5}{8} \end{bmatrix} r(t)$$

Soln

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_3 \\ -\frac{1}{3}x_1 - \frac{5}{3}x_2 - \frac{3}{4}x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{5}{8}r(t) \end{bmatrix}$$

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

$$\dot{x}_3 = -\frac{1}{3}x_1 - \frac{5}{3}x_2 - \frac{3}{4}x_3 + \frac{5}{8}r(t)$$

Assume  $x_1 = c(t)$  then  $\dot{x}_1 = x_2 = \frac{d}{dt}c(t)$

$$x_3 = \dot{x}_2 = \frac{d^2}{dt^2}c(t)$$

$$\dot{x}_3 = \frac{d^3}{dt^3}c(t)$$

$$\dot{x}_3 = -\frac{1}{3}x_1 - \frac{5}{3}x_2 - \frac{3}{4}x_3 + \frac{5}{8}r(t)$$

$$\frac{d^3 c(t)}{dt^3} = -\frac{1}{3}c(t) - \frac{5}{3}\frac{d}{dt}c(t) - \frac{3}{4}\frac{d^2}{dt^2}c(t) + \frac{5}{8}r(t)$$

$$\boxed{\frac{d^3 c(t)}{dt^3} + \frac{3}{4}\frac{d^2 c(t)}{dt^2} + \frac{5}{3}\frac{d c(t)}{dt} + \frac{1}{3}c(t) = \frac{5}{8}r(t)}$$

### ③ Signal flow Graph approach (SF9 state models).

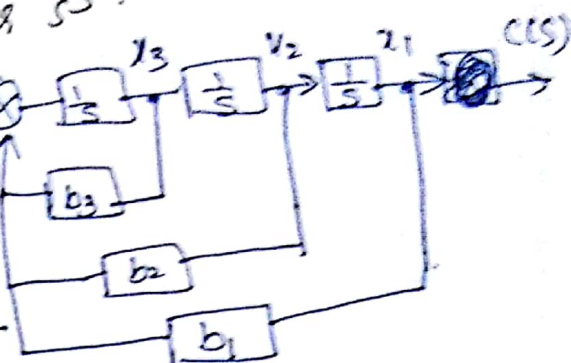
④

$$\text{Let } \frac{C(s)}{R(s)} = \frac{b_0}{s^3 + b_3 s^2 + b_2 s + b_1}$$

÷ both  $N_r$  &  $D_r$  by the highest power  $s^3$ .

$$\frac{C(s)}{R(s)} = \frac{b_0/s^3}{1 + \frac{b_3}{s} + \frac{b_2}{s^2} + \frac{b_1}{s^3}}$$

The denominator can be written as -

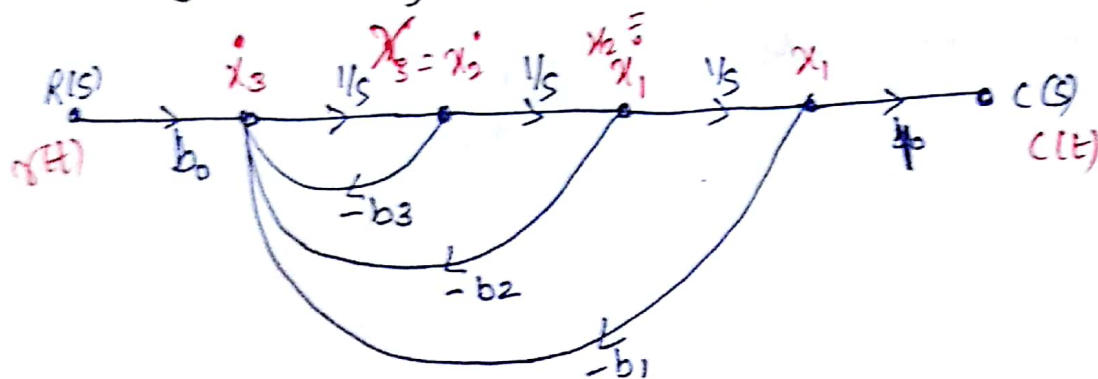


$$1 - \left( -\frac{b_3}{s} - \frac{b_2}{s^2} - \frac{b_1}{s^3} \right) = \Delta$$

$$\Delta = 1 - (L_1 + L_2 + L_3) \quad \text{where } L_1, L_2, L_3 \text{ are the loop gains}$$

Gains

$$L_1 = -\frac{b_3}{s} \quad L_2 = -\frac{b_2}{s^2} \quad L_3 = -\frac{b_1}{s^3} \quad \& \quad \text{forward path} = \frac{b_0}{s^3}$$



$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= b_0 r(t) - b_3 x_3 - b_2 x_2 - b_1 x_1 \end{aligned}$$

$$\text{o/p} \quad c(t) = x_1$$

$$y = [1 \ 0 \ 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Note: Assume the o/p of each integrator as a state.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -b_1 & -b_2 & -b_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ b_0 \end{bmatrix} r(t)$$



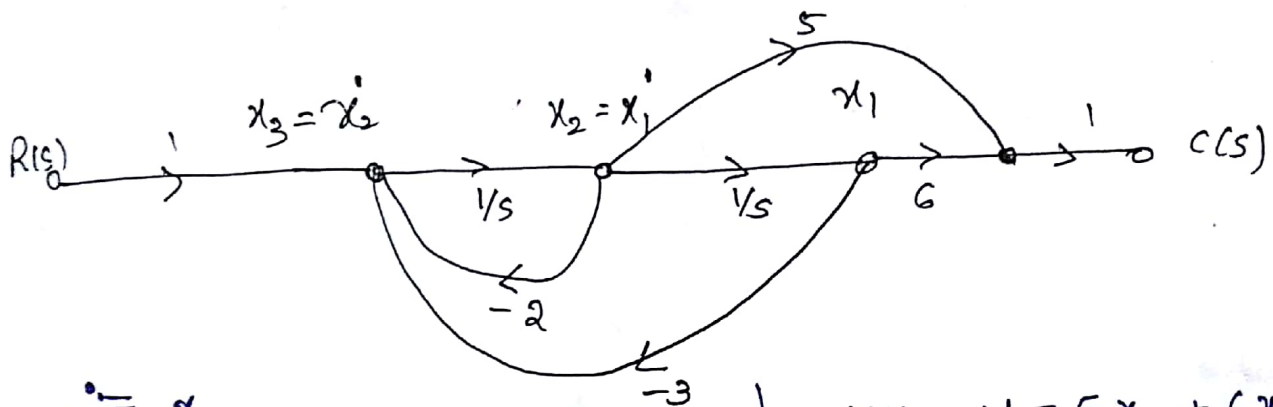
⑨ obtain the state model using SFG approach of system whose

TF is  $\frac{C(s)}{R(s)} = \frac{5s+6}{s^2+2s+3}$

÷ by  $s^2$

$$\frac{C(s)}{R(s)} = \frac{\frac{5}{s} + \frac{6}{s^2}}{1 + \frac{2}{s} + \frac{3}{s^2}} = \frac{\frac{5}{s} + \frac{6}{s^2}}{1 - (-\frac{2}{s} - \frac{3}{s^2})} = \frac{P_1 \Delta_1 + P_2 \Delta_2}{1 - (L_1 + L_2 + \dots)}$$

Thus there are 2 forward paths & 3 loops.



$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = r(t) - 2x_2 - 3x_1$$

$$C(t) = y = 5x_2 + 6x_1$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r(t) \quad \left| \quad y = \begin{bmatrix} 6 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

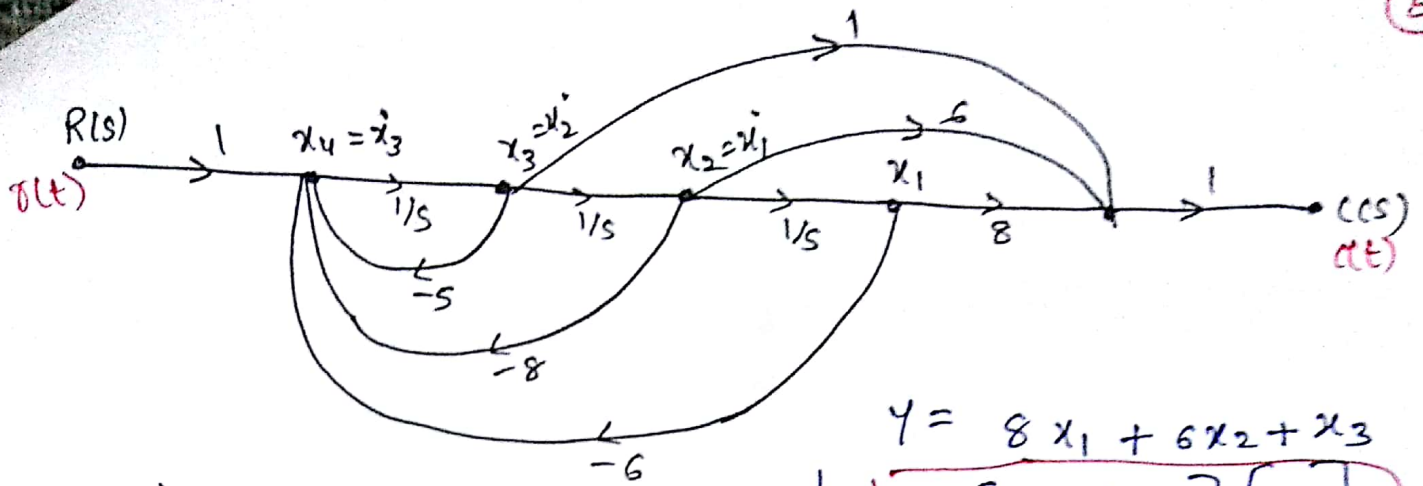
⑩  $\frac{C(s)}{R(s)} = \frac{s^2+6s+8}{s^3+5s^2+8s+6}$

÷ by  $s^3$

$$\frac{C(s)}{R(s)} = \frac{\frac{1}{s} + \frac{6}{s^2} + \frac{8}{s^3}}{1 - (-\frac{5}{s} - \frac{8}{s^2} - \frac{6}{s^3})}$$

3 forward paths & 3 loops

5



$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_3$$

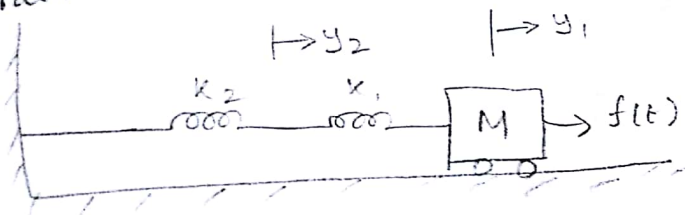
$$\dot{x}_3 = r(t) - 5x_3 - 8x_2 - 6x_1$$

$$y = 8x_1 + 6x_2 + x_3$$

$$y = \begin{bmatrix} 8 & 6 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -8 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} r(t)$$

Obtain the state-space representation of the following s/m.



$$\Rightarrow \left. \begin{aligned} M\ddot{y}_1 + k_1(y_1 - y_2) &= f(t) \\ k_2 y_2 + k_1(y_2 - y_1) &= 0 \end{aligned} \right\} \begin{array}{l} \text{--- (1)} \\ \text{--- (2)} \end{array}$$

$$x_1 = y_1 \Rightarrow \dot{x}_1 = \dot{y}_1$$

$$x_2 = y_2 \Rightarrow \dot{x}_2 = \dot{y}_2$$

$$\begin{aligned} \textcircled{1} \Rightarrow \quad & M\ddot{y}_1 = k_1(y_2 - y_1) + f(t) \\ & \ddot{y}_1 = \frac{k_1}{M}(y_2 - y_1) + \frac{f(t)}{M} \\ & \dot{x}_2 = \frac{k_1}{M}(y_2 - x_1) + \frac{f(t)}{M} \quad \text{--- (4)} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \Rightarrow \quad & (k_2 + k_1)y_2 - k_1 y_1 = 0 \\ & y_2 = \frac{k_1}{k_2 + k_1} \cdot x_1 \quad \text{--- (3)} \end{aligned}$$

sub. (3) in (4)

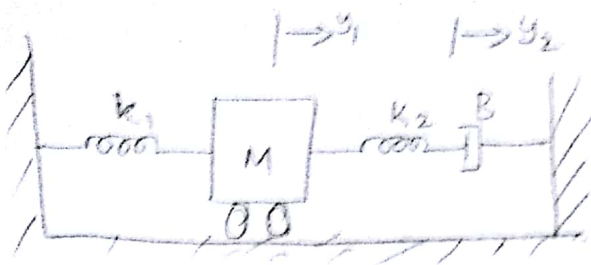
$$\dot{x}_2 = \frac{k_1}{M} \left( \frac{k_1}{k_2 + k_1} \right) x_1 - \frac{k_1}{M} x_1 + \frac{f(t)}{M}$$

$$\dot{x}_1 = x_2$$

$$\dot{\mathbf{x}} = \begin{pmatrix} \frac{k_1 k_1}{M(k_2 + k_1)} - \frac{k_1}{M} & 0 \\ 0 & 1 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 0 \\ \frac{1}{M} \end{pmatrix} f(t)$$

$$\mathbf{y} = (1 \ 0) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$





$$0 = M \ddot{y}_1 + k_1 y_1 + k_2 (y_1 - y_2) \quad \text{--- (1)}$$

$$0 = B \dot{y}_2 + k_2 (y_2 - y_1) \quad \text{--- (2)}$$

state variables,  $x_1 = y_1 \Rightarrow \dot{x}_1 = \dot{y}_1 = x_2$   
 $x_2 = \dot{y}_1 \Rightarrow \dot{x}_2 = \ddot{y}_1$   
 $x_3 = y_2 \Rightarrow \dot{x}_3 = \dot{y}_2$

$$\textcircled{1} \Rightarrow M \ddot{y}_1 = -k_1 y_1 + k_2 (y_2 - y_1)$$

$$\ddot{y}_1 = \dot{x}_2 = -\frac{k_1}{M} x_1 + \frac{k_2}{M} (x_3 - x_1)$$

$$\dot{x}_2 = \left( -\frac{k_1}{M} - \frac{k_2}{M} \right) x_1 + \frac{k_2}{M} x_3$$

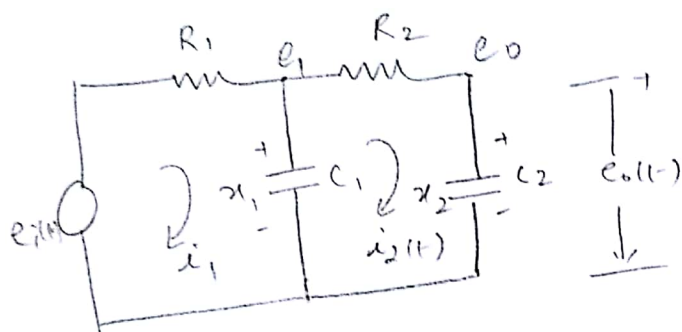
$$\textcircled{2} \Rightarrow B \dot{y}_2 = k_2 (y_1 - y_2)$$

$$\dot{y}_2 = \frac{k_2}{B} (y_1 - y_2)$$

$$\dot{x}_3 = \frac{k_2}{B} (x_1 - x_3)$$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -\frac{k_1}{M} - \frac{k_2}{M} & 0 & \frac{k_2}{M} \\ \frac{k_2}{B} & 0 & -\frac{k_2}{B} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$y = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$



$$x_1(t) = \frac{1}{C_1} \int_{-\infty}^t (i_1(\tau) - i_2(\tau)) d\tau \quad \text{diff.} \Rightarrow \frac{dx_1}{dt} = \frac{1}{C_1} (i_1 - i_2)$$

$$x_2(t) = \frac{1}{C_2} \int_{-\infty}^t i_2(\tau) d\tau \quad \Rightarrow \frac{dx_2}{dt} = \frac{1}{C_2} i_2$$

$$i_1 = \frac{e_i - e_1}{R_1} = \frac{e_i - x_1}{R_1} \quad \because e_1 = x_1$$

$$i_2 = \frac{e_1 - e_o}{R_2} = \frac{x_1 - e_o}{R_2} = \frac{x_1 - x_2}{R_2} \quad (\because e_o = x_2)$$

$$\Rightarrow \frac{dx_1}{dt} = \frac{1}{R_1 C_1} (e_i - x_1) - \frac{1}{R_2 C_1} (x_1 - x_2) = \frac{e_i}{R_1 C_1} - x_1 \left( \frac{1}{R_1 C_1} + \frac{1}{R_2 C_1} \right) + \frac{x_2}{R_2 C_1}$$

$$\Rightarrow \frac{dx_2}{dt} = \frac{1}{R_2 C_2} x_1 - \frac{1}{R_2 C_2} x_2 \quad \text{X } e_o = x_2.$$

$\Rightarrow$  no derivatives, 1st order DE

$$\begin{pmatrix} \frac{dx_1}{dt} \\ \frac{dx_2}{dt} \end{pmatrix} = \begin{pmatrix} \frac{-(R_1 + R_2)}{C_1 R_1 C_2} & \frac{1}{R_2 C_1} \\ \frac{1}{R_2 C_2} & \frac{-1}{R_2 C_2} \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} + \underbrace{\begin{pmatrix} \frac{1}{R_1 C_1} \\ 0 \end{pmatrix}}_B \frac{e_i}{u}$$

$$\dot{x} = A x + B u$$

$$\frac{e_o}{y} = \underbrace{(0 \quad 1)}_C \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$\begin{aligned} \dot{x} &= \underbrace{A}_{n \times n} x + \underbrace{B}_{n \times 1} u \quad \left\{ \begin{array}{l} \text{1st order DE} \rightarrow \text{easy to get solution} \\ \Rightarrow \text{state space representation} \end{array} \right. \\ y &= \underbrace{C}_{1 \times n} x \end{aligned}$$

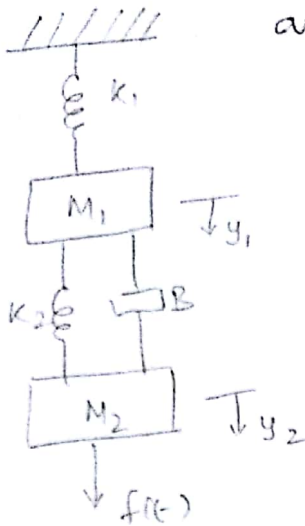
$A$  = system Matrix,  $B$  = input matrix,  $C$  = output matrix

$x_1$  &  $x_2$  = state variables.



23/01/14

For Mechanical s/m, the state variables are position & velocity.



For this system, since we have two

masses,  $x_1 = y_1$   $x_3 = y_2$   
 $x_2 = \dot{y}_1$   $x_4 = \dot{y}_2$

$\therefore$  for electrical s/m,  
 current through inductor  
 voltage through capacitor

$$f(t) = M_2 \ddot{y}_2 + B(\dot{y}_2 - \dot{y}_1) + k_2(y_2 - y_1) \quad \text{--- (1)}$$

$$0 = M_1 \ddot{y}_1 + B(\dot{y}_1 - \dot{y}_2) + k_2(y_1 - y_2) + k_1 y_1 \quad \text{--- (2)}$$

$$\dot{x}_1 = \dot{y}_1 = x_2$$

$$\dot{x}_2 = \ddot{y}_1 =$$

$$\dot{x}_3 = \dot{y}_2 = x_4$$

$$\dot{x}_4 = \ddot{y}_2$$

$$\textcircled{1} \rightarrow M_2 \dot{x}_4 = B(x_2 - x_4) + k_2(x_1 - x_3) + f(t)$$

$$\dot{x}_4 = \frac{B}{M_2} x_1 + \frac{B}{M_2} x_2 - \frac{k_2}{M_2} x_3 - \frac{B}{M_2} x_4 + \frac{f(t)}{M_2}$$

$$\textcircled{2} \Rightarrow M_1 \ddot{y}_1 = B(x_4 - x_2) + k_2(x_3 - x_1) - k_1 x_1$$

$$\dot{x}_2 = -\frac{B}{M_1}(k_2 + k_1) x_1 - \frac{B}{M_1} x_2 + \frac{k_2}{M_1} x_3 + \frac{B}{M_1} x_4$$

$$\begin{pmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -\frac{B(k_2 + k_1)}{M_1} & -\frac{B}{M_1} & \frac{k_2}{M_1} & \frac{B}{M_1} \\ 0 & 0 & 0 & 1 \\ \frac{k_2}{M_2} & \frac{B}{M_2} & -\frac{k_2}{M_2} & -\frac{B}{M_2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{M_2} \end{pmatrix} f(t)$$

$$\dot{x} = A x + B f(t) \rightarrow \text{state equation.}$$

12

$$\bar{y} = \begin{pmatrix} 1 & 0 & 0 & 0 \end{pmatrix}_{1 \times 4} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}_{4 \times 1} \quad y_1 - \text{o/p of interest.}$$

$\Rightarrow$  single i/p and single o/p s/m (SISO)

$$y = Cx, \quad C \rightarrow \text{o/p Matrix.}$$

Suppose,  $y_1$  &  $y_2$  are o/p of interest.

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}_{2 \times 4} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}_{4 \times 1} \quad \Rightarrow \text{SIMO}$$

In general,

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

where,  $D$  - feed through matrix

$$Y = CX + DU \longrightarrow \text{o/p Eq}^n$$

$$\dot{X} = AX + BU \longrightarrow \text{State Eq}^n .$$

Where

$X$  = State vector .

$\dot{X}$  = differential State vector .

$U$  = Input vector .

$Y$  = o/p . vector .

$A$  = System matrix .

$B$  = I/p matrix

$C$  = o/p matrix .

$D$  = feed-forward matrix .